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Publisher: Routledge

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The Service Industries Journal

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/fsij20>

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Version of record first published: 08 Jul 2011.

To cite this article: Riyad Eid & Hatem El-Gohary (2013): The impact of E-marketing use on small business enterprises' marketing success, The Service Industries Journal, 33:1, 31-50

To link to this article: <http://dx.doi.org/10.1080/02642069.2011.594878>

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The impact of E-marketing use on small business enterprises' marketing success

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(Received 29 March 2011; final version received 30 May 2011)

Small business enterprises (SBEs) are considered to be the economic engine leading to worldwide economic development. They have attracted substantial consideration from researchers, academics and practitioners in the last three decades. Meanwhile, E-marketing (EM) has emerged as one of the key drivers in sustaining an organisation's competitive advantage. Yet, there is a lack of systematic empirical evidence regarding marketing activities that are affected by the use of EM in the (SBEs) context, and their consequent performance outcomes. Therefore, the purpose of this paper is to examine the impact of EM use by SBEs on marketing success and to develop and test a conceptual model of the antecedents and consequences of EM use by SBEs. The conceptual framework consists of the following constructs: EM budget, EM tools, pre-sales activities, after-sales activities, marketing performance and marketing effectiveness. Moreover, 12 hypotheses were developed and tested. Exploratory and confirmatory factor analyses were used to test the validity of measures, while structural equation modelling was used in hypotheses testing. Data were collected from 114 SBEs who had used different EM tools. Findings reveal that the use of EM tools has a positive influence on SBEs pre-sales activities, after-sales activities, marketing performance and marketing effectiveness. The results of this study have major implications for the marketing domain, as they stress the central role of marketing people in the successful implementation of EM in SBEs.

Keywords: E-marketing; small business; marketing performance; marketing effectiveness

Introduction

Academic and managerial interest in E-marketing (EM) has been increasing in recent years. It is heralded by some as the new paradigm of marketing (see, e.g. Brodie, Winklhofer, Coviello, & Johnston, 2007; Eid, 2009; Eid & Trueman, 2004; Hotho & Champion, 2011; Wu, Mahajan, & Balasubramanian, 2003). The recent rush of publications in the area may give rise to the impression that EM can be applied in any context, yet there is little empirical evidence to support this.

Meanwhile, as small business enterprises (SBEs) are considered to be the economic engine leading worldwide economic development, they have attracted substantial consideration from researchers, academics and practitioners in the last three decades. A great deal of this interest derives from the belief that innovation, especially in information technology (IT), is crucially dependent on the potential of entrepreneurial SBEs. However, the recent

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revolution in computer science, the Internet, IT, media and communications has changed the nature of business and marketing practices. A growing numbers of companies and enterprises use the Internet and other electronic tools to communicate with suppliers, business customers and end users of their products and services. New forms of marketing have presented an opportunity for small businesses to grow in a dramatic and dynamic way.

The importance of SBEs is a reflection of their ability to create wealth and is based on their role as job providers. They bear the greatest responsibility of employment in the private sector all over the world. As a result, the development of small enterprises has been regarded as an important factor for the achievement of development objectives such as: poverty alleviation, economic development and the promotion of more democratic societies. In Europe, increasing attention has been given to the SBEs sector and to the contribution that entrepreneurs can make in transforming the European economy, especially in the current global economic climate (European Commission, 2009). This issue is reflected in the greater than ever range of European Union policy actions that are targeting SBEs.

Therefore, this research argues that the adoption of EM by the SBE can change the shape and nature of its business all over the world. The fast propagation of the Internet, the World Wide Web (WWW), ITs, communication technologies and computer sciences has created dynamic new electronic channels for marketing, and most companies today find it essential to have an online presence (Liang & Huang, 1998). But alongside these opportunities, there are problems associated with the dynamics of this new interactive media. These problems are exacerbated by the fact that much previous research has focused on the use of EM tools (e.g. the Internet) by large companies with the resources to adopt new technology to their specific needs rather than SBEs that have limited budgets and resources. Consequently, this research aims to add to the accumulative body of knowledge in the fields of EM and SBEs by focusing on investigating the impact of EM adoption on marketing success in the SBEs sector.

This study provides an insight for entrepreneurs, policymakers, practitioners, researchers and educators by providing a clearer view and deep understanding of the issues related to EM practices by SBEs as opposed to large companies. The research conceptual framework is developed keeping in mind that in most cases SBEs are on the disadvantaged end of the global digital world and might lose some benefits of EM (as the traditional literature suggests). Although the literature suggests that there are differences between EM use in SMEs and large companies, the current study focus only on investigating the impact of EM use on SBEs' marketing success.

Undoubtedly, this paper came to respond to these many calls for research in this area of SBEs. Despite the challenge presented to existing paradigms, the mainstream academic literature has largely ignored the growing importance of electronic-based marketing strategies. While numerous guides exist on 'How to do business' or 'How to make money' on the Internet, there have been few serious academic studies of the topic and little attempt has been made to develop conceptual frameworks for evaluating the effect of EM on SBEs marketing success. A major research initiative is required to improve our understanding in this area. In the absence of such an initiative, the mainstream academic literature will no longer accurately describe the reality of EM usage by SBEs. Therefore, this research overall aims to understand how the dynamics of EM have changed SBEs marketing practices and influenced their marketing performance.

Research questions and objectives

To analyse the implementation of the EM by SBEs, the researchers developed two major questions:

- (1) What progress have SBEs made towards EM implementation and where are they currently?
- (2) What is the relationship between EM adoption and marketing performance and marketing effectiveness of industrial and trading SBEs?

These were the overall questions to be answered by the current study; defined by the following three objectives:

- (1) to explore the current status of EM applications and practices in SBEs;
- (2) to develop and clarify a conceptual model integrating EM constructs, and its consequences on marketing performance and marketing effectiveness; and
- (3) to specify and test hypothesised relationships derived from the conceptual framework.

In the following sections, first the development of the conceptual model and the hypotheses of the study are presented. Next, the methodology of the study is discussed followed by the analysis and results. More specifically, the conceptual model is tested using path analysis, with the analysis of moment structures (AMOS) structural equation modelling package, and data collected by mail survey of 114 SBEs in the UK. Finally, the conclusions and their implications are discussed.

Literature review, conceptual model and hypothesised relationships

The conceptual model of this study is drawn from two streams of research: IT literature and the current EM theory. Figure 1 shows the conceptual model with the hypothesised linkages between the constructs. These linkages deal with three sets of hypotheses:

- (1) The effect of EM usage as expressed by the EM budget and EM tools, on SBEs marketing activities, as expressed by pre-sales activities and post-cost activities.

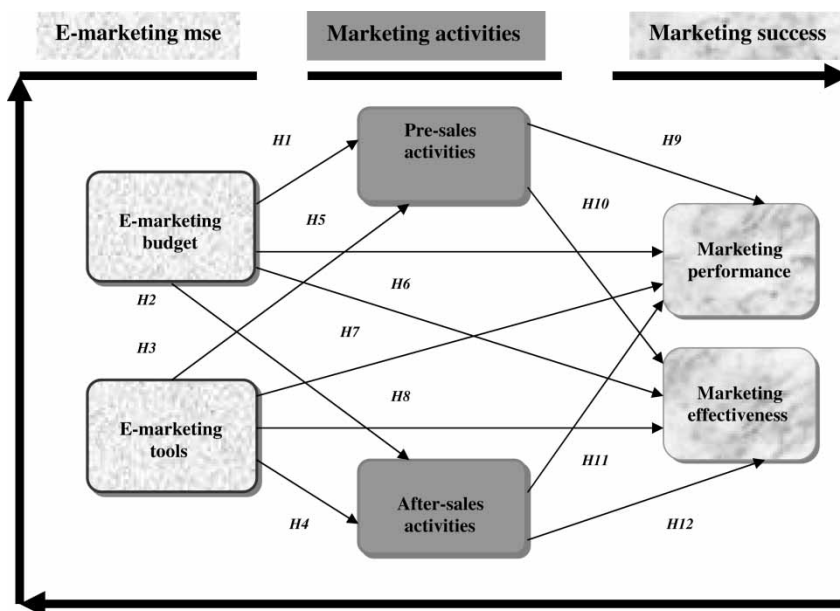


Figure 1. Proposed generic model for EM implementation.

- (2) The relationships between the marketing activities, as expressed by the pre-sales activities and post-cost activities, and the marketing success as expressed by the marketing performance and marketing effectiveness.
- (3) The effect of EM usage as expressed by the EM budget and EM tools, on marketing success as expressed by marketing performance and marketing effectiveness.

The next section provides a brief definition for each construct, followed by the development of the hypotheses. The relevant literature for each hypothesised relationship is discussed in the appropriate hypotheses development section.

Overview of the definition of SBEs

SBEs play a very important social and economic role in the UK, as they do in any other country all over the world. According to the statistics of the UK Department for Business, Enterprise and Regulatory Reform (BERR, 2009), the total number of registered business organisations in the UK at the beginning of 2007 (excluding government and non-profit organisations) was 4.7 million businesses, while almost all of these organisations (99.3%) being small (0–49 employees). Moreover, according to BERR (2009), SBEs accounted for 47.5% of employment and for 37.4% of the total turnover within the UK.

Regardless of the dominant position held by SBEs in today's economy, defining it has been a complicated task. There is very little agreement on what defines an SBE because the term covers a wide range of elements. Clearly, that there is no single unique definition of a small enterprise and this could be mainly because of their wide diversity. As a result, small businesses have been defined in a mystifying number of dissimilar ways according to the national and local needs of each country (Theng & Boon, 1996; Watson & Everett, 1996). This research has adopted the European definition for SBEs because it is relevant for this research; it is the legal definition in force within EU countries and it is an up-to-date definition that can take the dynamics of new technology into account. The definition was adopted by the European Commission in its Recommendation 2003/361/EC made on 6 May 2003 and was addressed to Member States, the European Investment Bank and the European Investment Fund (European Commission, 2009). The definition is made according to specific criteria which are: number of employees, annual turnover (or global balance) and independence.

Small businesses have a number of characteristics that are not shared with large businesses (e.g. lower levels of division of labour). Also SBEs tend to have similar characteristics such as being independently owned, close control exercised by owners, financially dependent on owners and critical decisions generally being made by owners (Australian Bureau of Statistics, 2001). From the point of view of many researchers and practitioners, the best illustration of the main important characteristics of a small enterprise remains that used by the Bolton Committee in its Report on Small Firms in 1971. The Committee described a small enterprise as an independent business managed by its owner or part-owners and has a small market share (DTI, 2008).

On the other hand, managerial characteristics of SBEs involve many things such as motivations, goals, objectives and actions of the owner and/or manager. Most of these managerial characteristics are highly related to the entrepreneur/owner as a manager. Deeks (1976) describes the small business manager/entrepreneur as a skilled craftsperson who is primarily concerned with both quality and reputation of his business. The importance of the entrepreneur/owner in the small business cannot be over-emphasised due to his/her role within the SBE which is not only a central role within the small business, but is also a great intellectual role for the business success.

EM usage

EM can be viewed as a new modern business practice and philosophy associated with buying and selling goods, services, information and ideas via the Internet and other electronic means. A review of relevant literature and published research revealed that the definitions of EM vary according to each researcher's point of view, background and specialisation. According to Strauss and Frost (2001), EM is defined as: 'The use of electronic data and applications for planning and executing the conception, distribution and pricing of ideas, goods and services to create exchanges that satisfy individual and organisational objectives' (Strauss & Frost, 2001, p. 454).

However, in this study, the use of EM usage is examined following Avlonitis and Karayanni (2000) who deal with it as a second-order construct that consist of two first-order components, EM budget and EM tools, captured by using one and five items, respectively. On one hand, the budget that is allocated to the EM may be used as an indicator of its usage. The relationship between marketing resources and performance has been a major area of interest in strategic management research over the last 20 years (Andersén, 2011). The classical form of allocating marketing resources usually looks for optimal allocation of marketing resources to marketing activities in order to maximise total profits (Albadvi & Koosha, 2011). On the other hand, EM tools include the use of any electronic data or electronic applications for conducting company marketing activities. As a result, EM includes Internet marketing, e-mail marketing, intranet marketing, extranet marketing, mobile marketing, telemarketing, electronic data interchange for marketing activities, customer relationship management and more. However, this research, and based on the results of reviewing the relevant literature, focuses on Internet marketing, e-mail marketing, intranet marketing, extranet marketing and mobile marketing (Chaffey, Ellis-Chadwick, Mayer, & Johnston, 2006; Eid & Trueman, 2004; El-Gohary, Trueman, & Fukukawa, 2008a, 2008b; Evans & King, 1999; Hofacker, 2001).

EM use and SBEs marketing activities

Undoubtedly, technology commercialisation is an important driver of a firm's marketing success (Ho, Fang, & Lin, 2011). The opportunities presented by EM for SBEs are considered as the Internet and other electronic media are now playing a vital role in the conducting of marketing activities by SBEs due to its unique characteristics both as a market and as a medium. A web site or a web page can have the potential to directly reach a large number of markets in a fast and economical way. With relatively low investments, almost any person who can read and write can have access to the WWW. EM provides SBEs with the opportunity of developing successful economic businesses in ways that have never been available to them before. It puts these entrepreneurs in touch with previously unavailable global resources and opportunities so that they can communicate and conduct business with new and existing customers in an integrated and easy way.

However, the review of the literature failed to find a single study that has been conducted to investigate the relationship between EM adoption and the marketing activities of SBEs. Consequently, this research expands the literature review to the broader concepts of E-Commerce, E-Business and to include other sizes of enterprises. Of the six studies identified through this extension, Domke-Damonte & Levsen, 2002, Garbi (2002), Khan and Motiwalla (2002), Wu et al. (2003), Drennan and McColl-Kennedy (2003) and Brodie et al. (2007), one study found a positive relationship between the EM and the marketing activities (Brodie et al., 2007) and five studies found a positive relationship between the E-Business penetration and firm performance

(Domke-Damonte & Levsen, 2002; Drennan & McColl-Kennedy, 2003; Garbi, 2002; Khan & Motiwalla, 2002; Wu et al., 2003). The most relevant studies of these six studies were the work of Domke-Damonte and Levsen (2002), Wu et al. (2003) and Brodie et al. (2007) as these studies investigated the relationship between adoption and marketing activities.

EM success

Generally speaking, there is no clear definition of a successful EM implementation. However, despite the importance of measuring EM success, there is little research on the measures used to evaluate the EM success. Within this context, Ambler and Kokkinaki (1997), based on an investigation for more than 1300 issues of seven marketing journals, found that only 11.5% of the investigated articles evaluated the marketing results. Furthermore, when looking to the EM success measures, it is noticed that there are many measures. In this respect, Clark (1998) identifies 16 measures, Ambler and Riley (2000) tested a total of 38 measures, Davidson (1999) considers 10 important measures of marketing effectiveness, and Meyer (1998) notes many other measures (Eusebio, Andreu, & Belbeze, 2006).

However, the results of the literature review suggest that a successful EM use is one that succeeds in meeting the business objectives. These objectives can be new sales, creating new customers, new markets, reduction of sales cost, increased profit, increased market share, increased brand equity any other objectives that are set by the organisation. These objectives have been classified into two main variables to measure the EM success in this study: namely, the marketing performance and marketing effectiveness.

Hypotheses

The relationship between the EM use and EM activities

The purpose of this research is to investigate the effects of the EM use as expressed by the EM budget and EM tools on SBEs marketing activities. Based on the previous literature, the effect of the EM use on marketing activities has been categorised into two basic dimensions: (1) pre-sales marketing activities and (2) after-sales marketing (Avlonitis & Karayanni, 2000). Figure 1 depicts the research model and illustrates the propositions tested in this study.

The first set of hypotheses examines the link between the use of EM as expressed by the EM budget and EM tools and SBEs marketing activities as expressed by SBEs pre-sales activities and after-sales activities. Many authors have argued that many pre-sales and after-sales marketing activities might be influenced by the use of EM (see, e.g. Avlonitis & Karayanni, 2000; Borders, Johnston, & Rigdon, 2001; Eid & Trueman, 2004; Furnell & Karweni, 1999; Honeycutt, Flaherty, & Benassi, 1998; Lancioni, Smith, & Oliva, 2000; Lord, 2001; Zhang & Duan, 2010).

A number of authors have paid attention to the consequences of the adoption of EM on SBEs pre-sales activities (Daniel & Wilson, 2002; Quayle, 2002; Simpson & Docherty, 2004; Vescovi, 2000; Wen, Chen, & Hwang, 2001). These consequences include faster discovery of customer needs, greater customisation of products, faster communication with customers and faster adaptability of customer needs. Other authors have argued that many after-sales marketing activities such as providing better service quality, developing new products, good customer relationships and increased customer satisfaction

might be influenced by the use of EM (see, e.g. Avlonitis & Karayanni, 2000; Borders et al., 2001; Eid, Elbeltagi, & Zairi, 2006; Eid & Trueman, 2004). So, it is hypothesised that:

H1: The higher the EM budget, the larger the impact on the pre-sales marketing activities.

H2: The higher the EM budget the larger the impact on the after-sales marketing activities.

H3: The use of the EM tools has a significant positive impact on the pre-sales marketing activities.

H4: The use of the EM tools has a significant positive impact on the after-sales marketing activities.

Effects of EM use on marketing success

The second part of the model investigates the relationships between EM use as expressed by the EM budget and EM tools and SBEs marketing success as expressed marketing performance and marketing effectiveness. Based on the literature, it was found that the EM adoption affects many issues that are related to the marketing performance and effectiveness of the SBEs such as: new sales (Simpson & Docherty, 2004; Walczuch, Van Braven, & Lundgren, 2000), new customers (Daniel & Wilson, 2002; Quayle, 2002), developing new markets and good customer relationships (Simpson & Docherty, 2004; Walczuch et al., 2000), improved productivity (Daniel & Wilson, 2002; Gunasekaran & Ngai, 2005; MacGregor & Vrazalic, 2004; Quayle, 2002; Rodgers, Yen, & Chou, 2002; Simpson & Docherty, 2004; Stockdale & Standing, 2004; Tsao, Lin, & Lin, 2004; Walczuch et al., 2000; Wen et al., 2001), increased market share (Eid & Elbeltagi, 2005), increased brand equity (Damanpour & Damanpour, 2001; Stockdale & Standing, 2004; Tsao et al., 2004; Wen et al., 2001) increased productivity (Daniel & Wilson, 2002; Gunasekaran & Ngai, 2005; Quayle, 2002; Rodgers et al., 2002; Simpson & Docherty, 2004; Stockdale & Standing, 2004; Walczuch et al., 2000; Wen et al., 2001).

EM adoption by SBEs can improve marketing performance and increase the marketing effectiveness through cost reduction resulting from the use of technology and EM tools (e.g. the Internet, e-mail, mobile phones, etc.) to carryout traditional marketing activities. This cost reduction resulted from the EM usage to improve the company profitability which will lead to better marketing effectiveness (Avlonitis & Karayanni, 2000; Borders et al., 2001; Furnell & Karweni, 1999; Honeycutt et al., 1998; Lancioni et al., 2000; Lord, 2001; Lynn, Lipp, Akgün, & Cortez, 2002). The following hypotheses are therefore proposed:

H5: The higher the EM budget the larger the impact on the marketing performance.

H6: The higher the EM budget the larger the impact on the marketing effectiveness.

H7: The use of the EM tools has a significant positive impact on the marketing performance.

H8: The use of the EM tools has a significant positive impact on the marketing effectiveness.

Effects of marketing activities on marketing success

The relationships between marketing activities as expressed by pre-sales marketing activities and post-sales marketing activities and marketing success as expressed by marketing performance and marketing effectiveness have been addressed in a number of studies (Daniel & Wilson, 2002; Eid & Elbeltagi, 2005; MacGregor & Vrazalic, 2004; Quayle, 2002; Rodgers et al., 2002; Simpson & Docherty, 2004; Stockdale & Standing, 2004; Tsao et al., 2004; Walczuch et al., 2000; Wen et al., 2001).

EM has been characterised as a tool for facilitating marketing efforts, thus leading to a higher level of marketing effectiveness (Anderson & Choobinen, 1996). Many effectiveness indicators have been cited in the literature as a result of EM adoption. These include

increased profits, increased market share, increased brand equity and increased productivity (Avlonitis & Karayanni, 2000; Borders et al., 2001; Eid et al., 2006; Furnell & Karweni, 1999; Honeycutt et al., 1998; Lancioni et al., 2000; Lord, 2001; Lynn et al., 2002). Accordingly, we put forward the following hypotheses:

H9: Pre-sales marketing activities have a significant positive impact on the marketing performance.

H10: Pre-sales marketing activities have a significant positive impact on the marketing effectiveness.

H11: After-sales marketing activities have a significant positive impact on the marketing performance.

H12: After-sales marketing activities have a significant positive impact on the marketing effectiveness.

Research methodology

Research design

This research aimed to develop a generic model for the effect of EM adoption by SBEs. After reviewing the literature, arguments are summarised into an integrated EM model, whose validity and value were tested by gathering data from 114 SBEs that utilise the different EM tools. Especially, based on the model, the study investigated the following:

the effect of EM use on SBEs marketing activities in terms of pre-sales and after-sales activities, and EM consequences on SBEs marketing success in terms of marketing performance and marketing effectiveness.

The sample

The survey questionnaire targeted a sample of 391 SBEs within the UK that had been selected randomly from a population of 1953 SBEs within the same region. As the study planned to obtain responses from different industries, so that generalisation of the findings could be established, the population were generated from some databases and business directories through searching the enterprises that are based in the UK and can satisfy the essential requirement to be considered as SBEs (number of employees and annual turnover). The following directories were used in generating the research population: E-Business Directory, Business Directory London, Internet Business Directory, Bizwiki, Freeindex, Countyweb, Business Directory UK, Alibaba Business Directory and FAME Business Directory. The sample was chosen to represent 20% of the population as accepted by most researchers within the field.

The sample size has been determined according to the Aaker and Day (1986) sample size equation, which is highly accepted by social science researchers since it takes into account the degree of required confidence, the sample error, ratio of population characteristics available in the sample (50% in social sciences) and population size. According to Aaker and Day (1986), the sample size can be determined depending on the following equation:

$$S = Z \sqrt{\frac{p(1-p)}{n}} \sqrt{\frac{N-n}{N-1}},$$

where Z is the degree of required confidence (95%), S the sample error (5%), P the ratio of population characteristics available in the sample (50%), N the population size and n the sample size.

Table 1. SBE survey response summary.

Total number of questionnaires	391
Number of completed and returned questionnaires	114
Unreachable SBEs	14
Number of SBEs declined participation	19
Response rate	31.84%

All the selected companies had implemented at least one of the EM techniques at least 3 year ago. A research packet, which contained a covering letter and an anonymous (self-administering) questionnaire, was mailed to the head of marketing departments (391 in total). This procedure resulted in 114 useful responses or a 31.84% overall response rate (Table 1). The response rate was calculated using the method proposed by De Vaus (1991, p. 99).

The sample can be described as follows: the majority of SBEs (78.9%) was located in England and only 4.4% of these SBEs were located in Northern Ireland. In addition, the majority of small businesses within the study was trading SBEs with a percentage of 64% of the total number of enterprises participating in the study and 36% of the participating enterprises were manufacturing SBEs. Moreover, the research sample was distributed among 11 different industries with the largest number of small businesses (29) in the computer and IT sector, representing 25.4% of the sample. With regard to the number of employees, it was found that the majority of SBEs (56.1%) falls into the category of enterprises that has between 10 and 19 employees. Moreover, 80.7% of the total number of enterprises had 39 employees or less. Meanwhile, the majority of SBEs within the study (21.1%) had less than 250,000 of annual sales. In addition, the majority of study SBEs (42.1%) had a marketing budget that is less than 10% of total enterprise budget and most of the research SBEs (29.8%) were in business for 11–20 years. Finally, it was found that the majority of the SBEs (28.1%) was in the category of less than 250,000 pounds as capital. on the other hand it was found that most of the research SBEs (64.9%) was working nationally.

To ensure that the valid responses were representatives of the larger population, a non-response bias test was used to compare the early and late respondents. Chi-square tests show no significant difference between the two groups of respondents at the 5% significance level, implying that a non-response bias is not a concern.

Research instrument development: measures

The development of the research instrument was based mainly on new scales, because we could not identify any past studies directly addressing all of the issues in this research. However, and where possible, we used validated measures that have been previously applied. All the constructs, with the exception of the one referring to the EM budget, included four items and were operationalised using five-point scales. Finally, we follow Avlonitis and Karayanni (2000) measuring the variable Internet budget by asking the respondents to indicate the percentage of their total marketing budget that the EM accounted for.

Two consecutive rounds of pre-testing were conducted in order to insure that respondents could understand the measurement scales used in the study: first, the questionnaire was reviewed by two academic researchers experienced in the questionnaire design and next, the questionnaire was piloted with four EM experts known to the researchers. The

pilot took the form of an interview, where the participants were first handed a copy of the questionnaire and asked to complete it followed by a discussion on any comments or questions they had. The outcome of the pre-testing process was a slight modification and an alteration of the existing scales, in light of the scales context under investigation. Based on the results and comments from the pilot tests, revisions were made to the questionnaire design.

Analysis and results

First, the psychometric properties of the constructs were assessed by calculating Cronbach's alpha reliability coefficient and the items-to-total correlation (Nunnally & Bernstein, 1994). These coefficients are represented for each of the constructs in Table 2. All scales have reliability coefficients ranging from 0.730 to 0.952, which exceed the cut-off level of 0.60 set for basic research (Nunnally, 1978).

Second, we performed an exploratory factor analysis (with Varimax rotation) to examine if the items for a construct share a single underlying factor (i.e. are unidimensional) to assess (a) SBEs marketing activities and (b) SBEs marketing success measures to produce a concise set of classification dimension. Items, which did not satisfy the following two criteria, were deleted: (1) dominant loadings greater than 0.5 and (2) cross-loadings less than 0.35 (Hair, Ralph, & Ronald, 1998).

The 16 items (variables) measuring the SBEs marketing activities in the research model were subjected to principal component factor analysis. Eigenvalues and the screen plot were used to determine the number of factors to be extracted. A four-factor structure was suggested using the criteria of an eigenvalue greater than 1 and the extracted factors account for 76.433% of the total variance. All factor loadings are generally high, and the lowest loading is equal to 0.537, while the Kaiser–Meyer–Olkin test of the factor analysis is substantial [0.908]. The resulting factor loadings are shown in Table 3. All items loaded onto the expected factors as they were originally designed. Factor loading were all higher than 0.5 on its own factors and, therefore, each item loaded higher on its associated construct than on any other construct. This supported the discriminant validity of the measurement.

Next, as suggested by Gerbing and Anderson (1988), tests for the unidimensionality of scales were performed, using confirmatory factor analysis (CFA) involving a single factor representation of each set of cogenerated items. Several fit statistics were utilised to evaluate the acceptability of each of the factor models. As recommended by Bentler and Bonnet (1980), the goodness-of-fit index (GFI) was utilised and deemed acceptable if above the recommended value of 0.90. Additionally, the comparative fit index (CFI) was also

Table 2. Measure of constructs' reliability.

Constructs	Number of items	Alpha	Mean	SD
EM tools*	4	0.774	2.155	0.626
Pre-sales factors [†]	4	0.819	4.192	0.927
After-sales factors [†]	4	0.940	4.142	0.834
SBEs marketing performance [‡]	4	0.952	3.850	0.672
SBEs marketing effectiveness [‡]	4	0.730	3.868	0.616

*EM uses.

[†]SBEs marketing activities.

[‡]SBE marketing success.

Table 3. Results of factor analysis.

	Component			
	Factor 1 Pre-sales activities	Factor 2 After-sales activities	Factor 3 Marketing performance	Factor 4 Marketing effectiveness
Faster discovery of customer needs	0.573			
Greater customisation of products	0.677			
Faster communication with customers	0.876			
Faster adaptability of customer needs	0.696			
Providing better service quality		0.835		
Developing new products		0.878		
Good customer relationships		0.768		
Increased customer satisfaction		0.718		
New sales			0.893	
New customers			0.864	
New markets			0.661	
Reduction of sales costs			0.641	
Increased profits				0.681
Increased market share				0.770
Increased brand equity				0.537
Increased productivity				0.899
Initial eigenvalues	9.073	1.295	1.091	1.011
% of variance	56.707	8.093	6.816	4.817
Cumulative%	56.707	64.800	71.616	76.433

used and the acceptable model fit is demonstrated with CFIs above 0.90, as well. Furthermore, the adjusted goodness-of-fit index (AGFI) and root mean square residual (RMSEA) also were provided. Standard cut-offs for the above indices, as proposed by experts (Hu & Bentler, 1995; Joreskog & Sorbom, 1982), are provided in Table 4. The results indicated that the scales were unidimensional.

We then the assessed convergent validity based on the results of the CFAs. The goodness-of-fit statistics indicate the unidimensionality of the measures (Anderson & Gerbing, 1988). All factor loadings were highly significant ($P < 0.001$) and all the estimates for the average variance extracted (AVE) were higher than the 0.50 level. Without exception, the

Table 4. CFA of model constructs.

Construct	Chi-square	<i>df</i>	<i>P</i>	GFI	AGFI	CFI	RMSEA
Use of EM tools	8.094	2	0.111	0.921	0.901	0.904	0.040
Pre-sales activities	2.365	2	0.307	0.989	0.947	0.998	0.040
After-sales activities	2.780	2	0.249	0.989	0.943	0.997	0.059
Marketing performance	6.468	2	0.089	0.971	0.854	0.967	0.080
Marketing effectiveness	1.689	2	0.430	0.992	0.962	0.990	0.000
Statistic				Suggested			
GFI				≥0.90			
AGFI				≥0.80			
CFI				≥0.90			
RMSEA				≤0.10			
Chi-square significant				≥0.05			

achieved AVE for the different constructs exceeded the recommended level of 0.50 (EM tools (0.67), pre-sales activities innovation (0.86), after-sales activities (0.69), EM performance (0.83) and EM effectiveness (0.61)).

Finally, given that the purpose of the study is to test the hypothesised causal relationships among the constructs of the model, we used the structural equation modelling package of AMOS. Since the sample size of 114 cases is not sufficient to support, a structural equation model at the level of complete disaggregation of measured variables (by using the multiple measured variables as indicators for each construct), we used the factor scores as single item indicators and performed a path analysis, applying the maximum-likelihood estimates (MLE) method, following the guidelines suggested by Joreskog and Sorbom (1982).

However, there are some general guidelines that have been proposed by some researchers with regard to the suitable sample size to be used when using the structural equation modelling in data analysis. For example, Hair et al. (1998) suggest that a sample with a size of less than 100 is considered to be a small sample. They also suggest that a medium sample size is between 100 and 200, and a large sample size is more than 200. On the other hand, Garson (2009) suggests that a sample size has to be more than 100. Moreover, many researchers have used a sample size of around 100 to conduct research, using the structural equation modelling approach (e.g. Eid, 2007; Khong, 2005). Based on that, it is generally regarded that a sample size of 100 is the practical acceptable size for using structural equation modelling.

The application of the MLE method for estimating the model entails that the constructs should satisfy the criterion of multivariate normality (Bagozzi & Yi, 1988). Therefore, for all the constructs, tests of normality, namely skewness, kurtosis and Mahalanobis distance statistics (Bagozzi & Yi, 1988), were produced. These indicated no departure from normality. Thus, as normality was confirmed for all the constructs, we proceeded in using the MLE method to estimate the model. Figure 2 illustrates the path diagram for the causal model. It also presents the estimated standardised parameters for the causal paths, their levels of significance and the square multiple correlations for each construct.

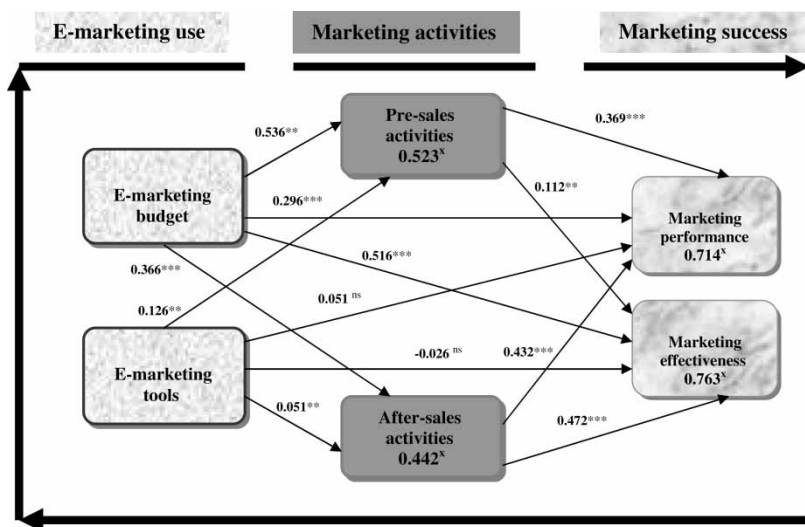


Figure 2. Results of path analysis.

A more detailed analysis of the results and measures for model fit are reported in Table 5. Since there is no definitive standard of fit, a variety of indices are provided along with the suggested guidelines. The chi-square statistic of the model was very small ($\chi^2 = 112.715$) and insignificant ($P = 0.216$), indicating a very good fit. Additionally, the results of the rest measures, together with the squared multiple correlations indicate that the overall fit of the model to the data is quite strong.

Since the causal effects of EM usage may be either direct or indirect, that is, mediated via the effects of other variables (pre-sales and after-sales marketing activities), or both, the total causal effects were computed. More specifically, the indirect effects are the multiplicative sum of the standardised path coefficients (Asher, 1983). The total effects are the sum of the direct effect and all the indirect effects. Table 6 shows the direct, indirect and total effects of the EM usage.

However, our findings generally support our conceptual model. The results place support to most of the hypotheses.

Table 5 shows the estimated standardised parameters for the causal paths. The EM budget positively affect all variables of the marketing activities, namely pre-sales marketing activities (*H1*) (standardised estimate = 0.536, $P < 0.05$) and after-sales marketing activities (*H2*) (standardised estimate = 0.366, $P < 0.01$). Similarly, EM tools positively affect the pre-sales marketing activities (*H3*) (standardised estimate = 0.126, $P < 0.05$) and the after-sales marketing activities (*H4*) (standardised estimate = 0.051, $P < 0.05$).

With respect to the SBEs' marketing performance, it was found that only three out of the four variables positively affect the SEBs marketing performance, namely the EM budget (standardised estimate = 0.296, $P < 0.01$), pre-sales marketing activities

Table 5. Standardised regression weights.

Predictor variables	Criterion variables	Hypothesised relationship	Standardised coefficient	$R^{2\ddagger}$
EM budget	Pre-sales activities	<i>H1</i>	0.536**	0.523
EM tools	Pre-sales activities	<i>H3</i>	0.126**	
EM budget	After-sales services	<i>H2</i>	0.366***	0.442
EM tools	After Sales Services	<i>H4</i>	0.051**	
EM budget	Marketing performance	<i>H5</i>	0.296***	0.714
EM tools	Marketing performance	<i>H7</i>	0.051 (ns)	
Pre-sales activities	Marketing performance	<i>H9</i>	0.369***	
After-sales services	Marketing performance	<i>H11</i>	0.432***	
EM budget	Marketing effectiveness	<i>H6</i>	0.516***	0.763
EM tools	Marketing effectiveness	<i>H8</i>	-0.026 (ns)	
Pre-sales activities	Marketing effectiveness	<i>H10</i>	0.112**	
After-sales services	Marketing effectiveness	<i>H12</i>	0.472***	
Statistic			Suggested	Obtained
Chi-square significance			≥ 0.05	0.216
GFI			≥ 0.90	0.943
AGFI			≥ 0.80	0.891
CFI			≥ 0.90	0.932
RMSEA			≤ 0.10	0.021

Note: ns is not significant.

$\ddagger$ This is the total variance explained in the referent dependent variable based on the hypothesised model.

** $P < 0.05$.

*** $P < 0.01$.

Table 6. Direct, indirect and total effect of the EM usage.

Criterion variable	Predictor variables	Direct effect	Indirect effect	Total effect
Pre-sales activities	EM budget	0.536	0.000	0.536
	EM tools	0.126	0.000	0.126
After-sales activities	EM budget	0.366	0.000	0.366
	EM tools	0.051	0.000	0.051
Marketing performance	EM budget	0.296	0.356	0.652
	EM tools	0.051	0.069	0.120
	Pre-sales activities	0.369	0.000	0.369
	After-sales activities	0.432	0.000	0.432
Marketing effectiveness	EM budget	0.516	0.233	0.749
	EM tools	-0.026	0.038	0.012
	Pre-sales activities	0.112	0.000	0.112
	After-sales activities	0.472	0.000	0.472

(standardised estimate = 0.229, $P < 0.01$) and the *after-sales marketing activities* (standardised estimate = 0.140, $P < 0.01$), have a significant and a positive effect upon *SBEs marketing performance*, supporting the hypotheses *H5*, *H9* and *H11*.

Similar to the SBEs' marketing performance, it was found that only three out of the four variables positively affect the SBEs' marketing effectiveness, namely the *EM budget* (standardised estimate = 0.516, $P < 0.01$), *pre-sales marketing activities* (standardised estimate = 0.112, $P < 0.01$) and *after-sales marketing activities* (standardised estimate = 0.472, $P < 0.01$), have a significant and a positive effect upon *SBEs marketing performance*, supporting the hypotheses *H6*, *H10* and *H12*.

Finally, SBEs' marketing performance (standardised estimate = 0.051, $P > 0.10$) and SBEs' marketing effectiveness (standardised estimate = -0.028, $P > 0.10$) are not directly affected by the *EM tools*. Thus, the results do not provide support for *H7* and *H8*. However, this insignificant direct effect of the *EM tools* on SBEs' marketing performance and the negative and insignificant direct effect of the *EM tools* on the SBEs' marketing effectiveness are strengthened by the indirect positive effect of the *EM tools* on SBEs' marketing performance and marketing effectiveness. This result may be interpreted by the fact that using *EM tools* does not guarantee marketing success since any potential advantage can only be gained by strategic planning and skilful use of the *EM tools* (Samiee, 1998). These findings support Avlonitis and Karayanni, (2000) and Eid et al. (2006) views that the mere use of the *EM tools* does not automatically lead to the marketing success. Therefore, it is not the use of *EM tools per se*, but rather the efforts of marketing staff that lead to successful *EM implementation* in terms of the SBEs' marketing performance and marketing effectiveness. The *EM tools* indirectly affect *marketing success* through the improvement of SBEs' pre-sales and post-sales activities. Indeed, the results indicate that marketing staff have a positive impact on the SBEs' *marketing success*.

Discussion and implications

The purpose of this article is (a) to offer some useful and practical guidelines for SBEs and other types of businesses wishing to successfully apply *EM tools* and (b) to enhance our understanding of its impact on the SBEs' marketing success.

EM use

EM usage is a measure of company's actual usage of the EM technology and has been measured using two variables: EM budget and EM tools. First, EM can be implemented through many different tools, the most common tools used by most firms are Internet marketing and e-mail marketing followed by mobile marketing and extranet marketing. All these EM tools were used by the research SBEs and based on the data analysis it was found that when implementing EM, SBEs depend on more than one tool of EM tools. Within this context, all the research respondents (114 SBEs with a percentage of 100% of the total) used Internet marketing as an EM tool. In addition 94.7% of the respondents used e-mail marketing as an EM tool and 36.8% of the respondents used mobile marketing. This is consistent with the findings of Ghosh (1998), Lancioni et al. (2000) and Eid (2009) who found that Internet marketing is the most commonly used tool for conducting marketing, electronically.

Marketing activities

The data suggest two types of marketing activities to be affected by the EM implementation. These types include pre-sales marketing activities and after-sales marketing activities. Pre-sales marketing activities have been measured using some indicators, namely faster discovery of customer needs, greater customisation of products, faster communication with customers and faster adaptability of customer needs. Similarly, after-sales marketing activities have been measured using some indicators, namely providing better service quality, developing new products, good customer relationships and increased customer satisfaction.

Overall, the EM usage variables (EM budget and EM tools) explain 52.3% of the *pre-sales marketing activities* and 44.2% of the *after-sales marketing activities*. This is consistent with the findings of Domke-Damonte and Levsen (2002), Garbi (2002), Khan and Motiwalla (2002), Drennan and McColl-Kennedy (2003), Wu et al. (2003), Brodie et al. (2007) who found a positive relationship between the E-Business penetration and EM.

Marketing success

One of the main aims of this research is to identify the impact of EM adoption by SBEs on the marketing success of these enterprises. Marketing performance and marketing effectiveness have been used to measure the perceived impact of EM adoption on the marketing success. Marketing performance has been measured using some performance indicators namely gaining new sales, gaining new customers, gaining new markets and reduction of sales cost. Similarly, marketing effectiveness has been measured using some effectiveness indicators, namely increased profits, increased market share and increased brand equity and increased productivity.

The findings show that allocating a sufficient budget to the EM implementation allows companies to interact, respond and communicate more effectively with their customers. The results also clearly demonstrated that pre-sales marketing activities and after-sales marketing activities have a catalytic influence on SBEs' marketing success (marketing performance and marketing effectiveness). Overall, the EM budget, pre-sales marketing activities and after-sales marketing activities, explain 71.4% of the SBEs marketing performance and 76.3% of the SBEs' marketing effectiveness.

We were surprised, however, to find that EM tools have shown only a negligible impact on EM performance and a negligible negative impact on EM effectiveness. However, upon closer examination of our study, this should not have been unexpected. This negligible direct effect (0.051) is strengthened by the indirect positive effect (0.069) of the use of EM tools on marketing performance. Similarly, the negligible negative direct effect (-0.026) is offset by the indirect positive effect (0.038) of the use of the EM tools on marketing effectiveness. This supports the findings of Avlonitis and Karayanni (2000) and Eid and Elbeltagi (2005) that state that it is not the use of the EM *per se*, but rather the efforts of the marketing department through the use of the EM tools that lead to enhanced marketing efficiency. EM tools indirectly affect marketing success through its use in different pre-sales and after-sales marketing activities.

This research has theoretical (academic) and managerial (practical) implications. In terms of academic implications, as the theory in the field of EM is still in its infancy stage and is not yet well established. This study can be considered as a step towards theory-building in the field of EM and has brought to light a number of concepts for the practice of EM by SBEs. Moreover, this study is one of the first studies to validate empirically the relationship between EM adoption and marketing success among SBEs.

On the other hand, this study has potential for managerial (practical) applications in the usage of EM by SBEs. First, not only does this study provide an empirical evaluation of the most important factors affecting the marketing success in SBEs, but it also measures the importance of such factors. Based on the importance of these factors revealed from the findings of the study, small business owners and/or marketing managers will have a better understanding about the different factors affecting the marketing success within their small businesses which then can be used in planning and directing the future policies, plans and strategies of these organisations. This will in turn lead to a positive impact on the economy.

Second, the study proves that there is a positive impact for EM adoption and SBEs marketing performance. Consequently, practitioners can derive a better marketing performance by adopting and implementing EM within their enterprises or companies. However, this does not mean that competitive advantage, marketing efficiency or marketing effectiveness is automatically achieved with the adoption of EM. Such benefits of EM adoption are based on the proactive and knowledgeable use of EM forms and tools by marketing people within the enterprise which in turn will leverage the enterprise marketing efficiency and effectiveness.

The findings illustrated that Internet marketing and e-mail marketing are the most commonly used EM tools by SBEs and that using such tools have a positive impact on the SBE success. The findings also showed that there are no differences between EM tools used by SBEs conducting different activities.

Limitations and suggestions for future research

As with any study, there are certain limitations that should be recognised. First, we assessed SBEs' marketing success by using marketing performance and marketing effectiveness, while there is evidence that SBEs' marketing success is a much broader construct that includes marketing efficiency (Avlonitis & Karayanni, 2000; Eid & Elbeltagi, 2005), customer satisfaction (Eid et al., 2006; Eid & Trueman, 2004) and brand equity (Damanpour & Damanpour, 2001; Stockdale & Standing, 2004; Tsao et al., 2004; Wen et al., 2001). Second, an extra limitation of this study is associated with its reliance on the subjective, self-report and judgemental indicators to measure the research constructs

within the survey questionnaire. Although the usage of objective indicators (such as sales volume, EM sales volume, total cost, EM cost, profit levels, numbers of customers, etc.) might improve accuracy, such measures are considered to be very sensitive and difficult to obtain accurately by survey respondents in general and small business owners and/or managers in particular. Third, the data are cross-sectional in nature, and hence, it is not possible to determine causal relationships.

The direction for future research, which emerged from our findings, is to investigate the EM adoption by other sizes of enterprises (e.g. micro-enterprises, medium-sized enterprises or large companies) depending on the same proposed factors generated within this study. Similar studies could be carried out to investigate EM adoption by service SBEs depending on the same proposed factors generated within this study. Future research may choose to investigate EM adoption by SBEs in other countries depending on the same proposed factors generated within this study. Finally, different constructs could be tried to measure the SBEs' marketing success.

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Appendix. Constructs, Scale items and sources

Construct	Measures used to capture constructs	Source
EM budget	Percentage of marketing budget accounted for EM	Adopted from Avlonitis and Karayanni (2000)
EM tools	Internet marketing, e-mail marketing, mobile marketing, intranet marketing, extranet marketing	New scale based on Eid and Trueman (2004) and El-Gohary et al. (2008a, 2008b)
Pre-sales activities	Faster discovery of customer needs, greater customisation of products, faster communication with customers, faster adaptability of customer needs	New scale based on Avlonitis and Karayanni (2000) and Eid and Trueman (2004)
After-sales services	Providing better service quality, developing new products, good customer relationships, increased customer satisfaction	New scale based on Avlonitis and Karayanni (2000), Eid and Trueman (2004) and El-Gohary et al. (2008a, 2008b)
Marketing performance	New sales, new customers, new markets, reduction of sales costs	Adopted from Avlonitis and Karayanni (2000)
Marketing effectiveness	Increased profits, increased market share, increased brand equity, increased productivity	New scale based on Eid and Trueman (2004) and El-Gohary et al. (2008a, 2008b)